

Phase I/II Operating Characteristics

Non-TITE alpha=0: individual-risk versus no-individual-risk

C2/C3 | continuous and IPDE-first enrollment | 56-day arrivals

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
OBD selection (%)						
C2 continuous individual risk one-stage	9	5.4	2.6	0.9	0.6	81.5
C2 continuous no individual risk one-stage	9.1	5.4	2.5	1	0.6	81.4
C3 continuous individual risk one-stage	9.1	5.5	3.1	0.9	0.4	81
C3 continuous no individual risk one-stage	9.6	5.8	3.1	0.9	0.4	80.2
C2 IPDE-first individual risk one-stage	8.3	6.3	2.6	1.6	0.2	81
C2 IPDE-first no individual risk one-stage	8.2	6.2	3	1.6	0.2	80.8
C3 IPDE-first individual risk one-stage	9.4	6.7	3.3	1.3	0.2	79.1
C3 IPDE-first no individual risk one-stage	9.5	7.4	3.6	1.1	0.2	78.2
C2 continuous individual risk two-stage	8.9	6	3	0.8	0.4	80.9
C2 continuous no individual risk two-stage	8.7	6.1	2.8	0.8	0.4	81.2
C3 continuous individual risk two-stage	9.3	6.3	3.2	0.9	0.4	79.9
C3 continuous no individual risk two-stage	9.4	6.2	3.3	0.9	0.6	79.6
C2 IPDE-first individual risk two-stage	8.5	6.1	2.7	1.6	0.2	80.9
C2 IPDE-first no individual risk two-stage	8.8	6.1	3.2	1.8	0.2	79.9
C3 IPDE-first individual risk two-stage	9.8	7.4	3.2	1.2	0.3	78.1
C3 IPDE-first no individual risk two-stage	9.8	7.4	3.5	1.2	0.2	77.9
Mean administrations by dose						
C2 continuous individual risk one-stage	11	4.3	2.4	1	0.3	
C2 continuous no individual risk one-stage	11	4.3	2.4	1	0.3	
C3 continuous individual risk one-stage	10.9	4.4	2.5	1	0.3	
C3 continuous no individual risk one-stage	10.9	4.4	2.5	1	0.3	
C2 IPDE-first individual risk one-stage	10.6	4.7	2.7	1.1	0.4	
C2 IPDE-first no individual risk one-stage	10.6	4.7	2.8	1.1	0.4	
C3 IPDE-first individual risk one-stage	10.8	4.7	2.6	1.1	0.4	
C3 IPDE-first no individual risk one-stage	10.8	4.7	2.6	1.2	0.4	
C2 continuous individual risk two-stage	10.9	4.4	2.5	1	0.3	
C2 continuous no individual risk two-stage	10.9	4.4	2.4	1.1	0.3	
C3 continuous individual risk two-stage	10.8	4.5	2.5	1	0.3	
C3 continuous no individual risk two-stage	10.9	4.4	2.5	1	0.4	
C2 IPDE-first individual risk two-stage	10.5	4.8	2.8	1.2	0.4	
C2 IPDE-first no individual risk two-stage	10.6	4.8	2.9	1.2	0.4	
C3 IPDE-first individual risk two-stage	10.8	4.8	2.6	1.1	0.4	
C3 IPDE-first no individual risk two-stage	10.7	4.8	2.7	1.2	0.4	

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C2/C3 | continuous and IPDE-first enrollment | 56-day arrivals

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
OBD selection (%)						
C2 continuous individual risk one-stage	22.8	22	19.1	21.3	13.8	1
C2 continuous no individual risk one-stage	23.4	21.5	19.4	21.6	13.2	0.9
C3 continuous individual risk one-stage	23.5	21.8	18.8	21.5	13.5	0.9
C3 continuous no individual risk one-stage	22	22.1	19.7	21	14.3	0.9
C2 IPDE-first individual risk one-stage	22.4	22.4	19.9	18	15.8	1.5
C2 IPDE-first no individual risk one-stage	23.4	21.8	19.2	19	15.2	1.4
C3 IPDE-first individual risk one-stage	23.9	22.1	18.3	19	15.7	1
C3 IPDE-first no individual risk one-stage	24.1	22.2	19.4	18.9	14.4	1
C2 continuous individual risk two-stage	26.1	19.8	18	20.7	14.4	1
C2 continuous no individual risk two-stage	24.6	21	18.4	21.2	13.8	1
C3 continuous individual risk two-stage	24.2	21.2	17.6	21.3	14.8	0.9
C3 continuous no individual risk two-stage	23.5	21.4	18.1	21.6	14.4	1
C2 IPDE-first individual risk two-stage	22	22	19.4	18.9	16.3	1.4
C2 IPDE-first no individual risk two-stage	21.6	22.5	19	19.3	16.1	1.5
C3 IPDE-first individual risk two-stage	23.1	22.5	18.5	19.7	15.2	1
C3 IPDE-first no individual risk two-stage	23.4	22.5	18.7	19.8	14.5	1.1
Mean administrations by dose						
C2 continuous individual risk one-stage	7.2	6.7	5.8	5.7	4.7	
C2 continuous no individual risk one-stage	7.3	6.6	5.7	5.8	4.6	
C3 continuous individual risk one-stage	7.3	6.6	5.6	5.8	4.7	
C3 continuous no individual risk one-stage	7.3	6.7	5.6	5.8	4.6	
C2 IPDE-first individual risk one-stage	7.1	6.5	5.9	5.5	5	
C2 IPDE-first no individual risk one-stage	7.2	6.5	5.9	5.5	5	
C3 IPDE-first individual risk one-stage	7.1	6.4	5.7	5.9	4.9	
C3 IPDE-first no individual risk one-stage	7	6.4	5.8	5.8	4.9	
C2 continuous individual risk two-stage	7.1	6.5	5.7	5.7	4.9	
C2 continuous no individual risk two-stage	7.2	6.6	5.7	5.8	4.9	
C3 continuous individual risk two-stage	7.1	6.6	5.6	5.8	4.8	
C3 continuous no individual risk two-stage	7.1	6.6	5.6	5.9	4.9	
C2 IPDE-first individual risk two-stage	6.9	6.4	6	5.7	5.1	
C2 IPDE-first no individual risk two-stage	6.9	6.4	6	5.7	5	
C3 IPDE-first individual risk two-stage	6.9	6.4	5.7	6	5.1	
C3 IPDE-first no individual risk two-stage	6.9	6.4	5.7	6	5.1	

Phase I/II Operating Characteristics

Non-TITE alpha=0: individual-risk versus no-individual-risk

C2/C3 | continuous and IPDE-first enrollment | 56-day arrivals

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
OBD selection (%)						
C2 continuous individual risk one-stage	5.3	45	29.7	12.8	2.2	5
C2 continuous no individual risk one-stage	5	46.6	29.2	12.2	2	5
C3 continuous individual risk one-stage	5.4	44.9	29.8	12.4	1.8	5.7
C3 continuous no individual risk one-stage	5.4	44.9	29.2	13	2.4	5.1
C2 IPDE-first individual risk one-stage	4.1	42.8	31.7	12.7	3.1	5.6
C2 IPDE-first no individual risk one-stage	4.7	44.1	30.8	12	3.1	5.3
C3 IPDE-first individual risk one-stage	4	46.3	28.6	13.7	2.4	5
C3 IPDE-first no individual risk one-stage	3.8	45.9	28.9	13.8	2.3	5.3
C2 continuous individual risk two-stage	5.2	46.7	29.1	12.1	2.2	4.7
C2 continuous no individual risk two-stage	5.2	47.8	28.7	11.5	1.9	4.9
C3 continuous individual risk two-stage	6	46.8	28.4	12.3	2.1	4.4
C3 continuous no individual risk two-stage	5	47.1	28.6	12	2.4	4.9
C2 IPDE-first individual risk two-stage	5.1	43.8	30.8	12.5	2.8	5
C2 IPDE-first no individual risk two-stage	4.8	45.5	30.3	11.9	2.5	5
C3 IPDE-first individual risk two-stage	4.2	45.4	29.9	12.9	2.3	5.3
C3 IPDE-first no individual risk two-stage	4.9	47.7	27.4	12.6	2.2	5.2
Mean administrations by dose						
C2 continuous individual risk one-stage	5.2	10.9	7.7	4.3	1.4	
C2 continuous no individual risk one-stage	5.2	11	7.7	4.3	1.4	
C3 continuous individual risk one-stage	5.2	10.9	7.6	4.4	1.5	
C3 continuous no individual risk one-stage	5.2	10.8	7.7	4.4	1.5	
C2 IPDE-first individual risk one-stage	5.2	10.6	7.9	4.3	1.6	
C2 IPDE-first no individual risk one-stage	5.2	10.7	7.8	4.3	1.7	
C3 IPDE-first individual risk one-stage	5.2	10.6	7.7	4.4	1.7	
C3 IPDE-first no individual risk one-stage	5.2	10.6	7.8	4.4	1.7	
C2 continuous individual risk two-stage	5	10.4	7.9	4.6	1.6	
C2 continuous no individual risk two-stage	5.1	10.3	7.9	4.6	1.6	
C3 continuous individual risk two-stage	5	10.4	7.7	4.8	1.7	
C3 continuous no individual risk two-stage	5	10.3	7.8	4.7	1.7	
C2 IPDE-first individual risk two-stage	5	10.1	8	4.7	1.9	
C2 IPDE-first no individual risk two-stage	5	10.1	8	4.7	1.9	
C3 IPDE-first individual risk two-stage	5.1	10.3	7.9	4.5	1.9	
C3 IPDE-first no individual risk two-stage	5.1	10.2	7.9	4.5	1.9	

Sources: C2/C3 alpha=0 non-TITE rows (56-day arrival) with individual risk and no individual risk.

Individual-risk rows use standard source files; no-individual-risk rows use files labeled noind. No OBD is reported from No_OBD_Selection_pct.

Phase I/II Operating Characteristics

Non-TITE alpha=0: individual-risk versus no-individual-risk

C2/C3 | continuous and IPDE-first enrollment | 56-day arrivals

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
OBD selection (%)						
C2 continuous individual risk one-stage	1.1	1.4	4.7	59.6	33.2	0
C2 continuous no individual risk one-stage	1.1	1.3	4.9	58.9	33.8	0
C3 continuous individual risk one-stage	1.2	1.1	4.9	59.7	33.1	0
C3 continuous no individual risk one-stage	1	1	4.5	59.2	34.3	0
C2 IPDE-first individual risk one-stage	1	1.3	4.7	59.8	33.2	0
C2 IPDE-first no individual risk one-stage	1	1.4	5.1	58.5	33.9	0.1
C3 IPDE-first individual risk one-stage	0.5	1.8	7	57.3	33.4	0
C3 IPDE-first no individual risk one-stage	0.7	1.9	6.1	58.2	33.1	0
C2 continuous individual risk two-stage	1.5	1.4	5.4	55.9	35.8	0
C2 continuous no individual risk two-stage	1.4	1.2	5.5	56	35.9	0
C3 continuous individual risk two-stage	1.2	1.3	4.5	55.9	37.1	0
C3 continuous no individual risk two-stage	1.3	1.3	4.7	56.1	36.6	0
C2 IPDE-first individual risk two-stage	0.8	1.1	5.7	55.8	36.5	0.1
C2 IPDE-first no individual risk two-stage	1	1.2	5.2	55.6	36.9	0.1
C3 IPDE-first individual risk two-stage	0.3	2.1	5.4	58.3	33.9	0
C3 IPDE-first no individual risk two-stage	0.3	2	5.6	59	33.1	0
Mean administrations by dose						
C2 continuous individual risk one-stage	3.4	3.5	4.4	12.2	6.5	
C2 continuous no individual risk one-stage	3.4	3.5	4.4	12.2	6.5	
C3 continuous individual risk one-stage	3.4	3.5	4.3	12.4	6.4	
C3 continuous no individual risk one-stage	3.4	3.4	4.3	12.5	6.4	
C2 IPDE-first individual risk one-stage	3.5	3.4	4.5	12.2	6.4	
C2 IPDE-first no individual risk one-stage	3.4	3.4	4.5	12.1	6.5	
C3 IPDE-first individual risk one-stage	3.4	3.5	4.5	12	6.6	
C3 IPDE-first no individual risk one-stage	3.4	3.5	4.5	12.1	6.5	
C2 continuous individual risk two-stage	3.4	3.4	4.3	10.7	8.2	
C2 continuous no individual risk two-stage	3.4	3.4	4.3	10.7	8.2	
C3 continuous individual risk two-stage	3.4	3.5	4.2	10.8	8.1	
C3 continuous no individual risk two-stage	3.4	3.4	4.2	10.8	8.1	
C2 IPDE-first individual risk two-stage	3.4	3.4	4.6	10.6	8	
C2 IPDE-first no individual risk two-stage	3.4	3.4	4.5	10.6	8	
C3 IPDE-first individual risk two-stage	3.4	3.5	4.3	11.2	7.7	
C3 IPDE-first no individual risk two-stage	3.4	3.5	4.2	11.2	7.7	

Phase I/II Operating Characteristics

Non-TITE alpha=0: individual-risk versus no-individual-risk

C2/C3 | continuous and IPDE-first enrollment | 56-day arrivals

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
OBD selection (%)						
C2 continuous individual risk one-stage	10.9	21.9	46.5	12	0.7	8
C2 continuous no individual risk one-stage	11.9	21.7	46.4	11.4	0.6	8
C3 continuous individual risk one-stage	11	22.8	46.2	11.4	0.4	8.2
C3 continuous no individual risk one-stage	10	22.3	47.1	11.6	0.7	8.3
C2 IPDE-first individual risk one-stage	11.3	21.6	49.3	11	0.7	6.1
C2 IPDE-first no individual risk one-stage	10.8	21.7	49.2	11.5	0.6	6.2
C3 IPDE-first individual risk one-stage	10.7	23.5	46.5	11.7	1.3	6.3
C3 IPDE-first no individual risk one-stage	10.8	22.9	46.9	12	1.1	6.3
C2 continuous individual risk two-stage	10.8	23.3	45.9	11.2	1.1	7.7
C2 continuous no individual risk two-stage	11.5	21.1	46.7	11.4	1.2	8.1
C3 continuous individual risk two-stage	11	22.9	44.7	11.9	1.6	7.9
C3 continuous no individual risk two-stage	11.2	23.1	45.3	11.2	1.3	7.9
C2 IPDE-first individual risk two-stage	10.7	23.9	48.7	9.9	0.8	6
C2 IPDE-first no individual risk two-stage	9.9	24.6	49.2	9.4	0.6	6.3
C3 IPDE-first individual risk two-stage	10.1	23.5	48	11.1	0.8	6.5
C3 IPDE-first no individual risk two-stage	9.9	22.8	48.1	11.2	1.5	6.5
Mean administrations by dose						
C2 continuous individual risk one-stage	7.1	7.8	9.9	3.7	0.6	
C2 continuous no individual risk one-stage	7.1	7.7	10.1	3.6	0.7	
C3 continuous individual risk one-stage	7.1	7.8	10	3.7	0.7	
C3 continuous no individual risk one-stage	7.1	7.8	10	3.7	0.7	
C2 IPDE-first individual risk one-stage	7.1	7.8	10.3	3.5	0.6	
C2 IPDE-first no individual risk one-stage	7	7.8	10.3	3.6	0.6	
C3 IPDE-first individual risk one-stage	7	7.8	9.9	3.9	0.6	
C3 IPDE-first no individual risk one-stage	7	7.7	10	3.9	0.6	
C2 continuous individual risk two-stage	6.8	7.5	9.7	4.1	1.1	
C2 continuous no individual risk two-stage	6.8	7.5	9.7	4.1	1.1	
C3 continuous individual risk two-stage	6.9	7.5	9.6	4.1	1.1	
C3 continuous no individual risk two-stage	6.8	7.6	9.5	4.2	1.1	
C2 IPDE-first individual risk two-stage	6.8	7.7	9.9	4	1.1	
C2 IPDE-first no individual risk two-stage	6.8	7.7	9.9	4	1	
C3 IPDE-first individual risk two-stage	6.7	7.6	9.7	4.2	1	
C3 IPDE-first no individual risk two-stage	6.8	7.6	9.6	4.2	1.1	

Sources: C2/C3 alpha=0 non-TITE rows (56-day arrival) with individual risk and no individual risk.

Individual-risk rows use standard source files; no-individual-risk rows use files labeled noind. No OBD is reported from No_OBD_Selection_pct.

Phase I/II Operating Characteristics

Non-TITE alpha=0: individual-risk versus no-individual-risk

C2/C3 | continuous and IPDE-first enrollment | 56-day arrivals

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 38						
DLT rate	0.05	0.1	0.12	0.15	0.3	
Efficacy rate	0.05	0.08	0.12	0.15	0.5	
OBD selection (%)						
C2 continuous individual risk one-stage	3.5	5.8	8.5	12.1	65.6	4.5
C2 continuous no individual risk one-stage	3.7	6.6	7.9	12.5	65.1	4.2
C3 continuous individual risk one-stage	3.7	6.1	7.1	12.4	66.5	4.2
C3 continuous no individual risk one-stage	3.7	6.5	8.1	12.2	65.3	4.2
C2 IPDE-first individual risk one-stage	4.9	5.7	8.9	11.8	64.5	4.2
C2 IPDE-first no individual risk one-stage	4	6.2	10.1	11.5	64.4	3.8
C3 IPDE-first individual risk one-stage	4.5	5.7	8.8	12.3	65	3.7
C3 IPDE-first no individual risk one-stage	4.1	5.7	8	12.4	65.9	3.9
C2 continuous individual risk two-stage	4.7	7.3	8	12	64.2	3.8
C2 continuous no individual risk two-stage	4	7.3	7.7	11.6	65.2	4.2
C3 continuous individual risk two-stage	4.4	7.1	8	11.6	64.9	4
C3 continuous no individual risk two-stage	4.9	6.6	7.4	11.7	65.5	3.9
C2 IPDE-first individual risk two-stage	4.8	5.9	10	11.5	63.7	4.1
C2 IPDE-first no individual risk two-stage	4.6	6	9.6	12	63.3	4.5
C3 IPDE-first individual risk two-stage	4.9	5.8	7.7	12.4	65.6	3.6
C3 IPDE-first no individual risk two-stage	4.9	5.3	8.6	12.2	65.1	3.9
Mean administrations by dose						
C2 continuous individual risk one-stage	4.5	4.4	5	5.5	10.5	
C2 continuous no individual risk one-stage	4.5	4.4	5	5.4	10.5	
C3 continuous individual risk one-stage	4.5	4.4	4.9	5.5	10.5	
C3 continuous no individual risk one-stage	4.5	4.4	4.9	5.4	10.6	
C2 IPDE-first individual risk one-stage	4.7	4.3	5.3	5.4	10.2	
C2 IPDE-first no individual risk one-stage	4.7	4.3	5.2	5.4	10.2	
C3 IPDE-first individual risk one-stage	4.7	4.5	5	5.2	10.6	
C3 IPDE-first no individual risk one-stage	4.7	4.5	4.9	5.3	10.6	
C2 continuous individual risk two-stage	4.5	4.4	4.9	5.5	10.7	
C2 continuous no individual risk two-stage	4.5	4.4	4.9	5.5	10.6	
C3 continuous individual risk two-stage	4.5	4.4	4.9	5.4	10.7	
C3 continuous no individual risk two-stage	4.5	4.4	4.9	5.5	10.7	
C2 IPDE-first individual risk two-stage	4.7	4.3	5.2	5.4	10.4	
C2 IPDE-first no individual risk two-stage	4.7	4.3	5.2	5.4	10.3	
C3 IPDE-first individual risk two-stage	4.6	4.3	4.9	5.4	10.7	
C3 IPDE-first no individual risk two-stage	4.6	4.3	4.9	5.3	10.7	

Sources: C2/C3 alpha=0 non-TITE rows (56-day arrival) with individual risk and no individual risk.

Individual-risk rows use standard source files; no-individual-risk rows use files labeled noind. No OBD is reported from No_OBD_Selection_pct.

Mean Unique Patients by Method

Mean number of unique patients per simulated trial

Page 7 of 7

Scenario	1	16	20	24	27	38	Mean
C2 continuous individual risk one-stage	12.3	18.2	18.9	19.5	19.4	18.9	17.9
C2 continuous no individual risk one-stage	12.3	18.2	18.9	19.6	19.4	18.9	17.9
C3 continuous individual risk one-stage	10.5	14.6	15.8	16.5	16.6	15.6	14.9
C3 continuous no individual risk one-stage	10.5	14.6	15.8	16.5	16.6	15.6	14.9
C2 IPDE-first individual risk one-stage	12.7	18.2	19	19.5	19.5	18.9	17.9
C2 IPDE-first no individual risk one-stage	12.6	18.2	18.9	19.5	19.5	18.9	17.9
C3 IPDE-first individual risk one-stage	10.6	14.4	15.7	16.5	16.5	15.5	14.9
C3 IPDE-first no individual risk one-stage	10.6	14.4	15.7	16.6	16.5	15.5	14.9
C2 continuous individual risk two-stage	12.4	18.2	19	19.7	19.6	18.9	18
C2 continuous no individual risk two-stage	12.4	18.3	18.9	19.7	19.6	18.9	18
C3 continuous individual risk two-stage	10.6	14.7	15.9	16.7	16.9	15.6	15.1
C3 continuous no individual risk two-stage	10.6	14.7	15.9	16.7	16.9	15.6	15
C2 IPDE-first individual risk two-stage	12.7	18.2	19.1	19.6	19.7	18.9	18
C2 IPDE-first no individual risk two-stage	12.7	18.2	19.1	19.6	19.6	19	18
C3 IPDE-first individual risk two-stage	10.7	14.5	15.8	16.7	16.8	15.6	15
C3 IPDE-first no individual risk two-stage	10.7	14.5	15.8	16.8	16.7	15.6	15